

EDGE INTELLIGENCE LAB PRACTICE

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IMPLEMENTATION AND DESIGN PIPELINE FOR OBJECT DETECTION IN VIDEOS USING YOLO V8

[OUTPUT VIDEO LINK](#)

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[6]
✓ 4m ▶ from ultralytics import YOLO
import cv2
import os

# 1. Load YOLOv8 model
model = YOLO("yolov8n.pt")

# 2. Open input video
video_path = "/content/SampleVideo_LowQuality.mp4"
cap = cv2.VideoCapture(video_path)

# 3. Get video properties for saving
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# 4. Define VideoWriter to save the result
output_path = "/content/output_tracking.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v') # Codec for .mp4
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

print("Processing video... please wait.")

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    # YOLOv8 tracking (stream=True is more memory efficient)
    results = model.track(frame, persist=True, verbose=False)

    # Draw results
    annotated_frame = results[0].plot()

    # Write the frame to our output file
    out.write(annotated_frame)

# Release everything
cap.release()
```

```
# Write the frame to our output file
out.write(annotated_frame)

# Release everything
cap.release()
out.release()
print(f"Success! Video saved at: {output_path}")

... Processing video... please wait.
Success! Video saved at: /content/output_tracking.mp4
```

FLOW DIAGRAM:

